

When Should Actuaries Trust Neural Networks?

A Benchmark of Classical and Neural Mortality Models with a Hybrid LC-ResNet and a Practical Decision Framework

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Abstract

We benchmark 14 mortality forecasting implementations—5 classical structural models, 2 simple baselines, 6 neural architectures, and a proposed hybrid—on 8 countries from the Human Mortality Database (1950–2023). Rolling-origin validation uses log-rate and age-100-truncated actuarial metrics to study short histories, forecast horizons, age segments, and the COVID-19 shock. **LC-ResNet** couples a Poisson Lee–Carter skeleton with a neural residual correction shrunk by $e^{-\lambda h}$ at horizon h . Key findings are benchmark-specific: the age-specific random walk with drift has the lowest full-grid log-rate error at every horizon; recurrent κ_t replacements improve on Lee–Carter at short horizons but deteriorate sharply on 20-year actuarial metrics; frozen rates are most robust over the cumulative 2020–2022 shock window; and LC-ResNet has the lowest one-year error among structured models. On a 1,000-life French annuity portfolio valued from 2033 rates, the five-model provision spread is 6.26 million EUR and the longevity SCR varies by 7% across models. The results caution against treating model complexity as a substitute for objective-specific validation.

Keywords: Mortality Forecasting · Lee–Carter · Deep Learning · Longevity Risk · Model Risk · Solvency II · Benchmark

JEL: C45 · C52 · C53 · G22 · J11

1 Introduction

Longevity risk—the risk that policyholders live longer than expected—is a primary concern for life insurers and pension funds. Under the Solvency II standard formula, the longevity-risk stress is an instantaneous permanent 20% decrease in mortality rates ([European Insurance and Occupational Pensions Authority, 2026](#)). The choice of mortality model is therefore relevant to both best-estimate liabilities and capital.

The Lee–Carter family ([Lee and Carter, 1992](#)) has dominated mortality forecasting for three decades. Recent work has expanded the toolkit through neural networks, shrinkage, multi-task learning, Bayesian model averaging, and variational inference ([Richman and Wüthrich, 2021](#); [Barigou et al., 2023](#); [Li et al., 2025](#); [De Mori et al., 2025](#); [Andersson and Lindholm, 2026](#)). The remaining practical issue is not whether flexible models can forecast mortality, but when their added complexity improves the particular quantity an actuary needs.

This paper fills that gap with four contributions:

1. **A reproducible benchmark** of 14 implementations on 8 countries, evaluated with log-rate and actuarial metrics under rolling-origin validation (Sections 4–6).
2. **LC-ResNet**, a proposed hybrid coupling a Poisson Lee–Carter skeleton with a neural residual network under horizon-dependent shrinkage (Section 4.3).
3. **Objective-specific guidance** mapping history length, horizon, age segment, and transparency constraints to evidence from this benchmark (Section 7).

4. **An actuarial case study** pricing a French annuity portfolio and quantifying model risk in EUR under a Solvency II longevity shock (Section 8).

Figure 1 summarizes the benchmark pipeline.

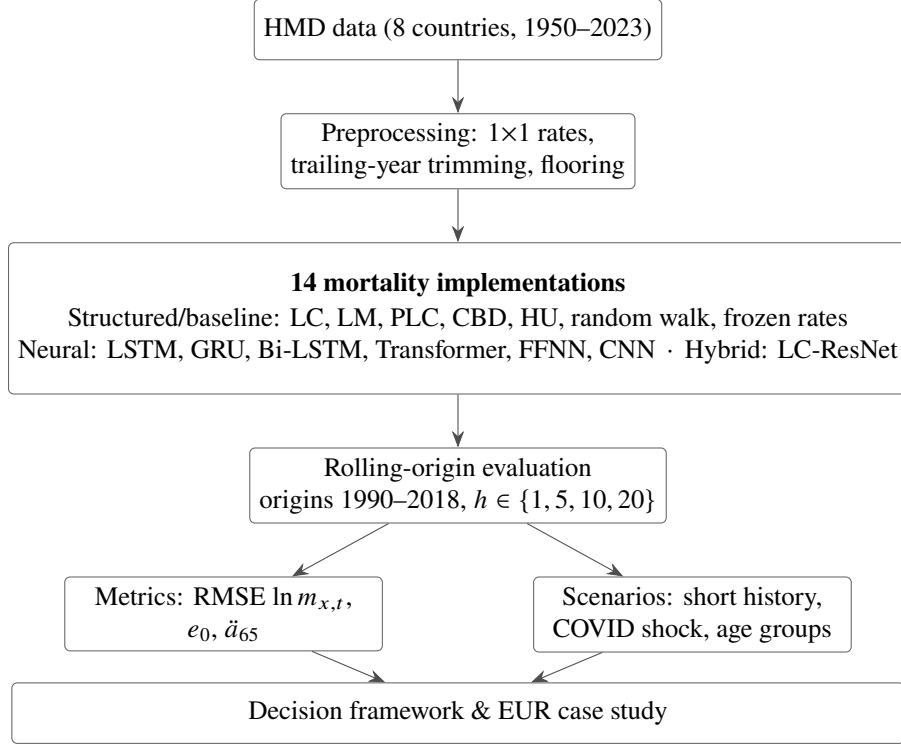


Figure 1: Benchmark pipeline: from raw HMD data to the decision framework and the actuarial case study.

2 Related Work

Lee and Carter (1992) introduced the seminal bilinear model for U.S. mortality. Extensions include the e_0 -adjusted variant of Lee and Miller (2001), fitting-period and jump-off refinements (Booth et al., 2002), the Poisson likelihood formulation of Brouhns et al. (2002), the two-factor CBD model for older ages (Cairns et al., 2006), and the functional data approach of Hyndman and Ullah (2007). Booth (2006) reviews this literature and documents the surprising strength of naïve extrapolation.

On the neural side, Nigri et al. (2019) replace the random-walk forecast of κ_t with an LSTM; Richman and Wüthrich (2021) propose a feed-forward network with entity embeddings; and Perla et al. (2021) study convolutional and recurrent architectures. More recent studies use shrinkage to combine domain models with machine learning (Li et al., 2025), share information through multi-task networks (De Mori et al., 2025), and increase age-time resolution (Zheng et al., 2026). Forecast combination and uncertainty are developed through leave-future-out Bayesian model averaging (Barigou et al., 2023) and variational inference (Andersson and Lindholm, 2026).

Our scope is deliberately narrower than several of these studies: single-population point forecasts on one harmonized data design. Within that scope, all implementations use the same rolling origins, neural training budget, and evaluation code; the comparison includes actuarial metrics and a EUR-denominated case study; and the proposed hybrid tests whether horizon shrinkage can bound a neural residual correction.

3 Data

We use 1×1 central death rates $m_{x,t}$ and exposures $E_{x,t}$ from the Human Mortality Database (HMD)¹ for eight countries: France, England & Wales, USA, Japan, Italy, Spain, Sweden, and the Netherlands. Ages span 0–100, years 1950–2023 (total population). The window includes the COVID-19 shock (2020–2023), used as a natural out-of-sample stress test.

The central death rate is defined as

$$m_{x,t} = \frac{D_{x,t}}{E_{x,t}}, \quad (1)$$

where $D_{x,t}$ denotes deaths at age x in year t and $E_{x,t}$ the matching central exposure. Zero or missing rates are floored at 10^{-6} before taking logarithms. Countries whose series end before 2023 (e.g. England & Wales, last available year 2022) have trailing missing years trimmed rather than imputed.

4 Models

The statistical models are implemented in Python (NumPy/SciPy), and neural models use PyTorch. Every implementation exposes a common `fit/forecast` interface, so the evaluation harness treats all 14 consistently, except that CBD is restricted to ages 60–100.

4.1 Classical models

Lee–Carter (LC). The bilinear structure

$$\ln m_{x,t} = a_x + b_x \kappa_t + \varepsilon_{x,t}, \quad (2)$$

estimated by SVD under the constraints $\sum_x b_x = 1$ and $\sum_t \kappa_t = 0$, with κ_t re-estimated on observed death counts. The index κ_t is forecast by a random walk with drift.

Lee–Miller (LM). Adjusts κ_t each year so that the model-implied life expectancy matches the observed e_0 (Lee and Miller, 2001).

Poisson Lee–Carter (PLC). Assumes $D_{x,t} \sim \text{Poisson}(E_{x,t} e^{a_x + b_x \kappa_t})$ and maximizes the likelihood by alternating Newton–Raphson updates (Brouhns et al., 2002).

Cairns–Blake–Dowd (CBD). For ages 60–100,

$$\text{logit } q_{x,t} = \kappa_t^{(1)} + \kappa_t^{(2)} (x - \bar{x}), \quad (3)$$

with a bivariate random walk with drift on $(\kappa_t^{(1)}, \kappa_t^{(2)})$ (Cairns et al., 2006).

Hyndman–Ullah-style functional model (HU). Penalized-spline smoothing of age profiles, functional principal components ($K = 6$), and random-walk-with-drift score forecasts. This is a simplified implementation of the broader robust functional procedure in Hyndman and Ullah (2007).

Baselines. An age-specific random walk with drift on $\ln m_{x,t}$, and frozen rates (last observed year held constant). Honest benchmarks require both.

4.2 Neural models

For benchmark fairness, every neural model uses the same budget: 200 epochs, early stopping with patience 20, Adam with learning rate 10^{-3} and weight decay 10^{-4} , fixed seed. Architectures are detailed in Appendix A.

Neural κ_t models. The LC skeleton (a_x, b_x) is retained; only the κ_t forecast is replaced by an LSTM, GRU, Bi-LSTM, or a two-layer Transformer encoder operating on a 20-year lookback window.

¹<https://www.mortality.org>. HMD data are free but not redistributable; the repository documents how to download them.

FFNN with embeddings. Following Richman and Wüthrich (2021): an age embedding concatenated with a continuous normalized year feature, mapped by a three-layer perceptron directly to $\ln m_{x,t}$. The continuous year encoding—rather than a year embedding—is what allows extrapolation to unseen future years.

CNN on the mortality surface. The (age $\times$ year) matrix is treated as an image; three convolutional layers predict the next annual profile from a sliding window, applied recursively for multi-step forecasts.

4.3 LC-ResNet: a hybrid with horizon shrinkage

Our proposed hybrid retains an interpretable LC skeleton and lets a small network correct what the skeleton misses:

1. Fit a Poisson Lee–Carter, obtaining $\hat{a}_x, \hat{b}_x, \hat{k}_t$.
2. Compute the structured residuals $r_{x,t} = \ln m_{x,t} - (\hat{a}_x + \hat{b}_x \hat{k}_t)$.
3. Train a residual network f_θ (MLP 64–32) on normalized features $(\tilde{x}, \tilde{t}, \tilde{b}_x, \tilde{k}_t)$ to predict $r_{x,t}$.
4. Forecast with horizon-dependent shrinkage:

$$\widehat{\ln m_{x,T+h}} = \underbrace{\hat{a}_x + \hat{b}_x \hat{k}_{T+h}}_{\text{LC skeleton}} + e^{-\lambda h} \underbrace{f_\theta(\tilde{x}, \tilde{t}, \tilde{b}_x, \tilde{k}_t)}_{\text{neural correction}}, \quad \lambda = 0.1. \quad (4)$$

At $h = 1$ the correction has weight $e^{-0.1} \approx 0.90$; at $h = 20$ its weight is $e^{-2} \approx 0.14$. The LC parameters remain available for reporting, but the residual correction is a black-box component; the complete hybrid should therefore be described as partially, not fully, interpretable.

5 Evaluation Protocol

Rolling-origin validation. For each origin $T_0 \in \{1990, 1992, \dots, 2018\}$, every model is re-trained from scratch on years $[1950, T_0]$ and evaluated at $h \in \{1, 5, 10, 20\}$ whenever the truth is available. A reported h -step error is calculated over the complete path $T_0+1, \dots, T_0+h$, not only the terminal year. Because country end dates differ, each table entry is the unweighted mean of the valid country–origin RMSE values.

Metrics. RMSE on $\ln m_{x,t}$; RMSE on period life expectancy at birth e_0 (years); and RMSE on the annuity-due factor $\ddot{a}_{65}$ at $i = 2\%$. Both actuarial quantities are valued on the observed 0–100 grid and are therefore truncated at age 100. They are reported only for full-grid forecasts; CBD is evaluated on log-rates over ages 60–100.

Scenarios. Three targeted experiments inform the guidance: (i) *short history*—one origin per country is chosen so that 10 evaluation years remain, then the training window is restricted to 20, 30, or 50 years and horizons 5 and 10 are averaged; (ii) *mortality shock*—training ends in 2019 and cumulative paths through 2020, 2021, and 2022 are evaluated; (iii) *age groups*—each model is refitted separately on ages 0–19, 20–64, and 65–100. We report point-accuracy comparisons only; formal predictive intervals and pairwise significance tests are outside this study.

6 Results

The benchmark yields 27,312 stored result rows across 14 implementations. Tables report unweighted means of country–origin RMSE values; the underlying country-level rows are in the repository. Pre-COVID rolling origins and the separate origin-2019 shock experiment are not pooled.

Table 1: RMSE on $\ln m_{x,t}$ by forecast horizon, averaged over 8 countries and all pre-COVID rolling origins. Lower is better; best full-grid model per column in bold. CBD forecasts ages 60–100 only, so its values are not comparable to full-grid models.

Model	$h=1$	$h=5$	$h=10$	$h=20$
<i>Classical</i>				
Lee–Carter	0.156	0.177	0.199	0.248
Lee–Miller	0.145	0.166	0.187	0.232
Poisson LC	0.190	0.214	0.231	0.272
CBD (60+ only)	0.076	0.087	0.095	0.120
Hyndman–Ullah	0.316	0.322	0.329	0.355
<i>Baselines</i>				
Random walk	0.100	0.124	0.146	0.203
Frozen rates	0.100	0.133	0.186	0.314
<i>Neural</i>				
LSTM κ_t	0.131	0.152	0.180	0.263
GRU κ_t	0.130	0.152	0.182	0.274
Bi-LSTM κ_t	0.132	0.157	0.195	0.288
Transformer κ_t	0.141	0.177	0.225	0.327
FFNN embeddings	0.231	0.257	0.290	0.367
CNN surface	0.147	0.191	0.240	0.448
<i>Hybrid</i>				
LC-ResNet	0.120	0.152	0.184	0.247

6.1 Accuracy on log death rates

Key findings.

1. The **age-specific random walk with drift dominates every full-grid structural model** on log-rate accuracy at every horizon (0.100 at $h=1$ to 0.203 at $h=20$)—a sobering baseline result consistent with Booth (2006). CBD is restricted to ages 60–100 and is not a full-grid competitor.
2. Among structured models, **LC-ResNet is the most accurate at $h=1$** (0.120) and ties the LSTM/GRU at $h=5$. At $h=20$, Lee–Miller is the strongest structured model (0.232), followed by LC-ResNet (0.247).
3. **Recurrent κ_t models** (LSTM, GRU) beat classical LC at $h \leq 10$ but lose that edge at $h=20$; the **Transformer and CNN degrade most** at long horizons (0.327 and 0.448), the classic extrapolation failure of flexible models.
4. **Frozen rates collapse with horizon** (0.100 $\rightarrow$ 0.314), as expected for a no-trend baseline: mortality does improve over 20 years.

6.2 Actuarial accuracy

The central observation is an **inversion between log-rate and actuarial accuracy for recurrent models**: the LSTM beats Lee–Carter on log-rates at $h \leq 10$ yet its 20-year e_0 error is 2.06 years, versus 0.69 for H-U and 0.74 for Lee–Miller. Small systematic biases can compound through a life table. LC-ResNet remains near the classical models, while the FFNN is unexpectedly strongest for the 20-year annuity metric. These differing winners motivate objective-specific model selection.

6.3 COVID-19 shock

Under the shock, the ranking flips: **frozen rates and several neural κ_t variants are the most robust full-grid forecasts**. COVID temporarily disrupted the improvement trends that LC-type models extrapolated.

Table 2: Actuarial metrics over cumulative 5- and 20-year paths: RMSE on age-100-truncated e_0 (years) and $\ddot{a}_{65}$. Unweighted mean across valid country–origin pairs; best per column in bold.

Model	RMSE e_0 (years)		RMSE $\ddot{a}_{65}$	
	$h=5$	$h=20$	$h=5$	$h=20$
Lee–Carter	0.428	0.832	0.178	0.397
Lee–Miller	0.318	0.742	0.195	0.398
Poisson LC	0.376	0.826	0.181	0.403
Hyndman–Ullah	0.419	0.694	0.188	0.437
Random walk	0.315	0.764	0.159	0.395
Frozen rates	0.607	2.417	0.272	1.096
LSTM κ_t	0.739	2.059	0.399	1.017
GRU κ_t	0.749	2.181	0.397	1.065
Bi-LSTM κ_t	0.882	2.375	0.459	1.149
Transformer κ_t	1.202	2.824	0.596	1.337
FFNN embeddings	0.557	0.787	0.316	0.381
CNN surface	0.591	1.047	0.292	0.490
LC-ResNet	0.353	0.846	0.174	0.403

Table 3: COVID-19 stress test, trained through 2019. Columns are cumulative forecast paths, not single-year errors. RMSE on $\ln m_{x,t}$; best full-grid model per column in bold.

Model	2020	2020– 21	2020– 22
Frozen rates	0.149	0.153	0.152
LSTM κ_t	0.150	0.156	0.158
Bi-LSTM κ_t	0.150	0.154	0.155
Transformer κ_t	0.150	0.154	0.153
GRU κ_t	0.153	0.161	0.166
LC-ResNet	0.152	0.167	0.183
Random walk	0.153	0.164	0.173
CBD (60+ only)	0.154	0.147	0.146
Lee–Miller	0.170	0.183	0.195
Lee–Carter	0.180	0.193	0.205
CNN surface	0.197	0.213	0.230
Poisson LC	0.236	0.252	0.265
FFNN embeddings	0.269	0.270	0.268
Hyndman–Ullah	0.336	0.337	0.337

This stress test covers one unusual episode, so it supports a narrow conclusion: calm-period ranking does not identify shock robustness.

6.4 Short history and age segments

With 20 training years, frozen rates have the lowest full-grid error (0.134), while LC-ResNet is the best structured model (0.159). On refitted age-band models, LC-ResNet and Bi-LSTM tie for the lowest young-age error after rounding (0.228). For ages 65–100, frozen rates lead (0.054), followed by H-U and the random walk (0.068); CBD records 0.102 in its intended age range. Full tables appear in Appendix B.

7 Decision Framework

Table 4 distills the results into benchmark-specific guidance. There is no universal winner: recommendations change with the target loss and with whether a structured mortality model is required.

Table 4: Which model to use when. Recommendations follow directly from the scenario results.

Context	Recommendation	Evidence
Full-grid log-rate accuracy	Random walk	Lowest error at every horizon in Table 1
Short history (≤ 25 yr)	Frozen rates; LC-ResNet if structured	Lowest overall and structured errors, respectively (Table 9)
Shock sensitivity analysis	Frozen rates plus model ensemble	Frozen rates lead this one COVID episode; retain alternatives
20-year e_0 accuracy	H-U; Lee–Miller as LC-family option	Errors 0.694 and 0.742 years, respectively
20-year annuity accuracy	FFNN, with classical sensitivity runs	Lowest $\ddot{a}_{65}$ error (0.381), narrowly ahead of random walk and LC
Short horizon, structured	LC-ResNet	Lowest structured error at $h=1$; tie at $h=5$
Elderly ages	Frozen rates / H-U / random walk	Lowest age-band errors; CBD is not the winner here
Transparent skeleton needed	LC family or LC-ResNet with caveat	LC parameters are reportable; LC-ResNet correction remains opaque

The table is not a universal prescribing rule. In practice, several plausible models should be retained for sensitivity analysis, and selection should be repeated on the exact population, loss function, and forecast horizon used in the liability calculation.

8 Actuarial Case Study

Setup. A stylized portfolio contains 1,000 French pensioners aged 65, each receiving 12,000 EUR annually. The five models are fitted to France through 2023 and the age-specific rates in their 10-year-ahead forecast (2033) are used with $i = 2\%$. Payments and survival are valued through age 100. For CBD, forecast rates replace ages 60–100 in the last full French age vector; ages below 65 do not enter the annuity value.

The provision spread across models reaches **6.26 MEUR** (2.8% of the largest provision), driven mainly by CBD, which projects higher old-age mortality; among full-grid models the spread is 1.22 MEUR. Either way, the differences are material at portfolio scale.

The longevity SCR ranges from 11.52 to 12.33 MEUR—a 7% relative spread—while the SCR-to-provision ratio ranges from 5.2% to 5.8%. This is an illustrative deterministic standard-formula calculation, not an insurer-specific valuation.

Table 5: Illustrative 2033 annuity pricing by model. The provision is $1,000 \times 12,000 \times \ddot{a}_{65}$, with payments truncated at age 100.

Model	$\ddot{a}_{65}$	Provision / person (EUR)	Total (MEUR)
Lee–Carter	18.338	220,050	220.05
Poisson LC	18.323	219,873	219.87
LC-ResNet	18.268	219,217	219.22
Random walk	18.236	218,826	218.83
CBD	17.816	213,793	213.79
Inter-model spread			6.26

Table 6: Solvency II standard-formula longevity shock: permanent 20% reduction in mortality rates (European Insurance and Occupational Pensions Authority, 2026).

Model	Base (MEUR)	Shocked (MEUR)	SCR (MEUR)	SCR (%)
Lee–Carter	220.05	231.57	11.52	5.23
Poisson LC	219.87	231.39	11.52	5.24
LC-ResNet	219.22	230.80	11.58	5.28
Random walk	218.83	230.54	11.71	5.35
CBD	213.79	226.12	12.33	5.77

9 Discussion

The random walk result. The strongest single finding is that an age-specific random walk with drift beats every structured model on log-rate accuracy at every horizon. Yet the winner changes under actuarial losses, showing why a benchmark should include both simple baselines and decision-relevant metrics.

When neural networks help, and how they fail. Recurrent κ_t models improve short-horizon log-rate accuracy and are competitive in the COVID experiment, but their long-horizon e_0 errors exceed two years. Conversely, the FFNN performs poorly on log-rate RMSE yet leads the 20-year annuity metric. Model complexity must be assessed against the decision-specific loss.

Bounded flexibility is promising but not dominant. LC-ResNet’s shrinking correction yields the lowest one-year structured-model error and classical-level 20-year actuarial errors. It ties, rather than uniquely wins, the young-age experiment and does not beat Lee–Miller at 20 years. The skeleton is transparent; the residual correction is not.

Limitations. All models are single-population; cohort effects are not explicit; the H-U implementation uses random-walk score forecasts rather than automatic ARIMA selection; neural results use one fixed seed and one shared training budget rather than model-specific tuning; predictive distributions and statistical significance are not evaluated; and life-table quantities are truncated at age 100. Unequal country end dates create unequal numbers of valid origins, and post-2023 observations are needed to assess post-COVID trends. The case study is stylized and its five-model spread is sensitive to the inclusion of age-restricted CBD.

10 Conclusion

When should actuaries trust neural networks for mortality forecasting? Only after validation against the exact target quantity. Here, neural κ_t models help at short horizons but fail on long-horizon actuarial metrics; a feed-forward model nevertheless leads the 20-year annuity metric; and the proposed LC-ResNet offers a useful structured compromise at short horizons. The random walk and frozen-rate baselines remain indispensable. The 6.26 MEUR five-model provision spread illustrates why model choice should be reported as a sensitivity, not hidden behind a single ranking.

Code Availability

The complete implementation—14 benchmarked models, evaluation harness, case-study script, tests, and interactive dashboard—is openly available at:

<https://github.com/aminemanai2003/neural-mortality-benchmark>

HMD data are not redistributable; the repository documents the download procedure. Benchmark rows are regenerated with `python scripts/run_benchmark.py`; the EUR tables are regenerated with `python scripts/run_case_study.py` (seed 42).

Use of Generative AI

OpenAI Codex was used for language editing, consistency checks across code and manuscript, and document quality assurance. The author remains responsible for the study design, implementation, results, citations, and final text.

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A Model Configurations

Table 7: Hyperparameters. All neural models: 200 epochs max, early stopping (patience 20), Adam, learning rate 10^{-3} , weight decay 10^{-4} , seed 42.

Model	Architecture
LSTM / GRU / Bi-LSTM κ_t	2 layers, hidden 64, lookback 20
Transformer κ_t	2 encoder layers, $d_{\text{model}}=64$, 4 heads, lookback 20
FFNN embeddings	age emb. dim 8 + continuous year; MLP 128–64–32, dropout 0.1
CNN surface	channels 16–32–64, kernel 3×3, adaptive pooling
LC-ResNet	residual MLP 64–32, dropout 0.1, $\lambda = 0.1$
CBD	age range 60–100
Hyndman–Ullah	6 functional components, penalized splines

Table 8: Mean training time per fit (seconds, single CPU).

Model	Time (s)	Model	Time (s)
Frozen rates	0.00	Bi-LSTM κ_t	0.47
Random walk	0.00	Transformer κ_t	0.55
CBD	0.00	GRU κ_t	0.73
Hyndman–Ullah	0.00	LC-ResNet	1.45
Lee–Carter	0.01	FFNN embeddings	3.27
Poisson LC	0.05	CNN surface	8.18
Lee–Miller	0.09		

B Additional Results

Table 9: Short-history scenario: RMSE on $\ln m_{x,t}$ with training truncated to the last 20 or 50 years (averaged over countries and horizons $h \in \{5, 10\}$). Best full-grid model in bold.

Model	20 yr	50 yr
CBD (60+ only)	0.128	0.122
Frozen rates	0.134	0.134
LC-ResNet	0.159	0.151
Poisson LC	0.163	0.189
Lee–Miller	0.164	0.163
Lee–Carter	0.165	0.163
LSTM κ_t	0.165	0.157
Bi-LSTM κ_t	0.165	0.147
GRU κ_t	0.165	0.156
Transformer κ_t	0.165	0.164
Random walk	0.173	0.149
CNN surface	0.178	0.220
FFNN embeddings	0.290	0.255
Hyndman–Ullah	0.340	0.333

Table 10: Age-group evaluation: RMSE on $\ln m_{x,t}$ for young (0–19) and elderly (65–100) ages, averaged over countries, origins, and horizons. Best per column in bold.

Model	Young 0–19	Elderly 65–100
LC-ResNet	0.228	0.070
Bi-LSTM κ_t	0.228	0.071
LSTM κ_t	0.229	0.074
GRU κ_t	0.230	0.074
Lee–Carter	0.233	0.079
Lee–Miller	0.236	0.080
Frozen rates	0.241	0.054
Transformer κ_t	0.246	0.078
FFNN embeddings	0.252	0.117
Random walk	0.263	0.068
Hyndman–Ullah	0.346	0.068
Poisson LC	0.314	0.073
CNN surface	0.332	0.125
CBD (60+ only)	–	0.102